

DRAFT

NPS National Water Supply Vulnerability Assessment

Technical Report

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Table of Contents

Plain-Language Summary: What This Tool Does and Why It Matters	3
1. Overview and Purpose	5
2. Study Area and Water Supply Systems	5
2.1 Geographic Scope	5
2.2 Water Supply Database.....	5
3. Vulnerability Framework.....	6
3.1 Conceptual Basis.....	6
3.2 Indicator Development Status	6
3.3 Aggregation: Euclidean Distance.....	7
4. Exposure Indicators.....	8
4.1 Demand Exposure Factor	9
4.2 Drought Exposure Factor.....	11
4.3 Coastal Impacts Exposure Factor [Semi-Provisional]	12
4.4 Temperature Exposure Factor [Provisional]	14
4.5 Water Supply Exposure Factor [Semi-Provisional]	15
4.6 Wildfire Exposure Factor.....	17
5. Sensitivity Indicators	18
5.1 Demand Sensitivity Factor [Semi-Provisional]	18
5.2 Drought Sensitivity Factor	19
5.3 Flood Sensitivity Factor.....	20
5.4 Sea Level Rise Sensitivity Factor	20
5.5 Source Type Sensitivity Factor.....	21
5.6 Temperature Sensitivity Factor [Provisional]	22
5.7 Water Supply Trend Sensitivity Factor	23
5.8 Wildfire Sensitivity Factor	24
6. Priority Classification.....	24
7. Contribution Calculations	25
8. Limitations and Caveats.....	26
8.1 Point Representation and Buffer Distance.....	26
8.2 Equal Weighting.....	26
8.3 Relative Scoring	27
8.4 Climate Scenario Uncertainty	27
Appendix A: Water Balance Model Variable Explorer.....	28

Variable Descriptions.....	29
Appendix B: Data Sources Summary.....	30
References	32

Plain-Language Summary: What This Tool Does and Why It Matters

National parks provide clean drinking water to millions of visitors and staff every year. That water comes from rivers, streams, springs, and wells, all of which can be affected by climate change. Rising temperatures, shifting precipitation, more frequent wildfires, rising seas, and more intense floods all have the potential to reduce the quantity or quality of water available to parks. This tool helps NPS staff and stakeholders answer a simple question: which park water supply systems are most at risk from climate change, and why?

How Does this Tool Define Vulnerability?

Think of vulnerability like the risk of getting sick during flu season. You are more at risk if (a) there is a lot of flu going around in your community (exposure), and (b) you have a weakened immune system or underlying condition (sensitivity). We apply the same logic to water supplies. A water system's vulnerability is determined by:

- **Exposure:** How much is climate projected to change the conditions around this water supply? Are runoff and precipitation projected to decline? Is wildfire risk increasing? Will sea levels rise nearby?
- **Sensitivity:** How susceptible is this system to those changes right now? Does it rely on a vulnerable water source type? Is visitation growing, increasing demand? Is it already in a flood zone or a fire-prone landscape?

This definition of vulnerability follows the National Park Service (NPS) planning approach and definitions (NPS 2021), which is based upon the internationally recognized vulnerability framework established by the fourth Intergovernmental Panel on Climate Change (IPCC 2007). An important modification of our work is that we do not consider the component of Adaptive Capacity. This component refers to a system's ability to adapt to change and we consider it only relevant for living resources, in agreement with other climate vulnerability assessments (Michalak et al. 2021, NPS 2016, Rockman et al. 2016). As this assessment is focused on water supplies, which inherently cannot adapt without human intervention, we view adaptive capacity irrelevant in this scope.

What Do the Scores Mean?

Every water supply receives an overall vulnerability score (or "index"), where higher numbers indicate greater vulnerability. Scores are shown as percentiles, representing where a water supply falls relative to others; for example, a score of 90 means the supply scores higher than 90% of other water supplies. **These scores are relative** - they show how a system compares to all other NPS water supplies within the **tool's current filtered state**. By default, the tool opens to national rankings, but users have the option to filter by NPS Region and state. At each of these filtering steps, the vulnerability score is recalculated on the filtered subset of water supplies.

What Impacts are Assessed?

Six primary impact pathways are assessed:

1. **Water Availability:** Trends, timing, and projected declines in source water (precipitation or runoff) that could reduce overall supply.
2. **Drought:** Trends and projected change in water deficit (SPEI), reflecting increasing drought conditions.

3. **Wildfire:** Current fire hazard and projected fire risk. Wildfires can contaminate watersheds, modify hydrology, and damage infrastructure.
4. **Flooding:** Current and future (*forthcoming*) risk to flooding events, indicating risk of infrastructure damage and source water contamination.
5. **Sea Level Rise:** Coastal inundation, saltwater intrusion, and storm surge for coastal park water systems.
6. **Demand:** Current and future water use trends reflecting the potential for increased water stress.

How Should Results Be Used?

This is a rapid screening tool, not a final decision-making instrument. This tool is designed with flexibility at the forefront, capable of serving many purposes for many different users. Results should be used to:

- Identify systems warranting more detailed, park-specific assessment.
- Identify potential hazards that warrant further assessment.
- Prioritize resources for climate adaptation planning.
- Communicate water supply risks to park managers, regional staff, and various other stakeholders.

1. Overview and Purpose

This report documents the methods underlying the National Park Service (NPS) National Water Supply Vulnerability Assessment, an interactive screening tool for evaluating climate change vulnerability of water supply systems managed by NPS across the contiguous United States (CONUS). The assessment is designed as a nationally consistent, repeatable screening tool that can be updated as new data becomes available and used to identify potential hazards to park water supplies and prioritize systems for more detailed vulnerability analyses.

Climate change poses significant risks to the hydrological systems that support NPS water supplies. Projected increases in temperature, shifts in precipitation seasonality and amount, more frequent and severe wildfires, accelerated sea level rise, and changes in flood frequency all have direct implications for water availability, water quality, and the integrity of water supply infrastructure (IPCC 2021, Vörösmarty et al. 2010). At the same time, growing visitor use and regional competition for water resources increase the sensitivity of many park water systems to these physical changes (Fisichelli et al. 2015).

The framework used in this assessment characterizes vulnerability through two primary components: Exposure - the degree to which a water supply is projected to experience climate-related change, and Sensitivity - current conditions and inherent susceptibility to those changes. This two-component structure follows the NPS planning approach, based upon the IPCC (2007) vulnerability framework, and is directly adapted from the methodological approach of Michalak et al. (2021, 2026), who developed a strategic climate change vulnerability assessment for national parks. Our indicators and scoring methodology have been adapted to better fit a facilities perspective for water supply risk. Adaptive capacity - the inherent ability of a system to cope with change - is only relevant for living systems; we therefore exclude it in our quantitative framework and instead focus on management actions that can mitigate identified vulnerabilities.

Key output: An overall vulnerability score (displayed as a percentile ranking) computed for each water supply system, alongside component scores (Exposure, Sensitivity), and individual indicators grouped into factors, including hazard-specific flags for wildfire, flood, sea level rise, and drought (Figure 1).

2. Study Area and Water Supply Systems

2.1 Geographic Scope

The assessment covers all NPS-managed water supply systems within CONUS. Water supplies in Alaska, Hawaii, and U.S. territories are excluded due to data availability constraints for many indicators.

2.2 Water Supply Database

Water supply records are drawn from the NPS Water Supply Database, an internal database of water supply source locations (latitude/longitude) associated with park facilities. The following filters and modifications are applied:

- Exclude systems with source type recorded as municipal or hauled - these supply types receive water from external utilities or delivery services outside NPS control. Users can, however, view these systems as independent layers on the map.
- For systems missing source-level coordinates ($n = 73$), the park unit centroid is substituted.
- Exclude all non-CONUS water supplies.

After filtering, 1,410 individual water supplies across 207 park units are retained.

3. Vulnerability Framework

3.1 Conceptual Basis

The framework follows the NPS definition for vulnerability assessments (NPS 2021), in which a system's vulnerability is a function of its exposure to climate-related changes and its sensitivity to those changes.

The indicator hierarchy consists of four levels: (1) individual indicators, (2) vulnerability factors (thematic groupings of indicators), (3) two top-level components (Exposure and Sensitivity), and (4) a total vulnerability index computed based on the Exposure and Sensitivity component scores (Table 1). See Section 3.3 for aggregation calculations and Section 8.3 for weighting analyses.

Vulnerability Scores are Relative

Because this tool is designed for a wide range of users, from single park managers to regional planners, the vulnerability framework is centered around a **relative approach**. By default, the tool displays the national vulnerability score, ranking all water supplies within our filtered CONUS dataset and producing their relative vulnerability score. However, whenever the user filters the map to an NPS Region or state, the index is re-calculated from the raw indicator scores of the filtered dataset.

3.2 Indicator Development Status

The indicators and factors described in Sections 4 and 5 (and in Table 1) are at different stages of development. Because the assessment is being reviewed by park partners while several indicators are still being refined, each indicator and factor carries a development status so that users and reviewers can tell how settled a given method is and what the corresponding scores can support. We report three levels of development status throughout:

Implemented. The methodology is stable and the indicator is calculated for all applicable water supplies and included in the scores presented in the tool. Results can be interpreted as described in the corresponding methods section.

Provisional. The indicator is calculated for all applicable water supplies and is included in the scores presented in the tool, but its methodology is still being revised. Scores for these indicators are expected to change, and the methods sections identify what is under revision. Because these indicators contribute to composite scores, revisions to them will also change factor, component, and total vulnerability scores.

Planned. The vulnerability pathway has been identified as relevant, but the methodology has not yet been determined, and the indicator is not calculated or included in any score presented in the tool.

Because scores are calculated from the indicators available at the time of this release, results should be understood as reflecting the currently implemented and provisional set rather than the full framework described here. Adding indicators changes both the

composite scores and the relative ranking of water supplies, so results presented in the tool should be viewed as preliminary.

Table 1. Summary of our indicator, factor and component groupings. Detailed descriptions of individual indicators can be seen in later sections.

Status	Indicator	Vulnerability Factor	Component
Implemented	Change in nearby water use	Demand	Exposure
Provisional	Change in supply and demand Mismatch*		
Implemented	Change in SPEI (Water Deficit)	Drought	
Planned	Change in Flood Risk	Flood	
Implemented	Inundation by Sea Level Rise	Coastal Impacts	
Provisional	Inundation by Storm Surge		
Implemented	Saltwater Intrusion		
Provisional	Change in Air Temperature	Temperature	
Planned	Water Quality	Water Quality	
Planned	Water Rights	Water Rights	
Implemented	Percent Change in Runoff or Precipitation*	Water Supply	
Provisional	Change in Runoff or Precipitation Timing*		
Implemented	Change in likelihood of wildfire	Wildfire	
Implemented	Historic Visitation Trend	Demand	Sensitivity
Implemented	Nearby water use		
Provisional	Current Supply and Demand Mismatch*		
Implemented	Historic Trend in SPEI (Water Deficit)	Drought	
Implemented	Current Flood Risk	Flood	
Implemented	Current Inundation	Sea Level Rise	
Implemented	Depth to Water Table	Source Type	
Provisional	Historic Trend in Air Temperature	Temperature	
Planned	Water Treatment Type	Water Quality	
Planned	Water Rights	Water Rights	
Implemented	Historic Trend in Runoff or Precipitation*	Water Supply	
Implemented	Wildfire Hazard Potential	Wildfire	

* These components will use either runoff, precipitation, or an “unlimited” source (e.g., ocean supplies) depending on the water supply type.

3.3 Aggregation: Euclidean Distance

Indicators are aggregated at each hierarchical level using Euclidean distance from a reference point of zero vulnerability, following Michalak et al. (2021, 2026) (Figure 1). Each indicator forms one axis of a multidimensional stressor space, and a site's score is its displacement from an origin representing no vulnerability on any axis. Because the distance is not divided by the number of indicators, a low score on one indicator does not dilute a high score on another as it would under simple averaging; because indicator values are squared, the aggregate is weighted toward the most

severe individual stressor rather than toward the number of stressors present. This makes Euclidean distance a partially compensatory aggregation in the sense of OECD/JRC (2008), sitting between the fully compensatory arithmetic mean and a non-compensatory maximum, and alternative schemes such as the geometric mean would instead reward co-occurrence of moderate stressors and produce different rankings.

Separately, because the total vulnerability score is calculated across all factor scores simultaneously rather than as a two-stage roll-up of the Exposure and Sensitivity components, a low score on one component cannot mask a high score on the other. All factors enter this calculation with equal weight. A sensitivity analysis testing whether that choice materially affects scores and priority classifications is reported in Section 8.3.

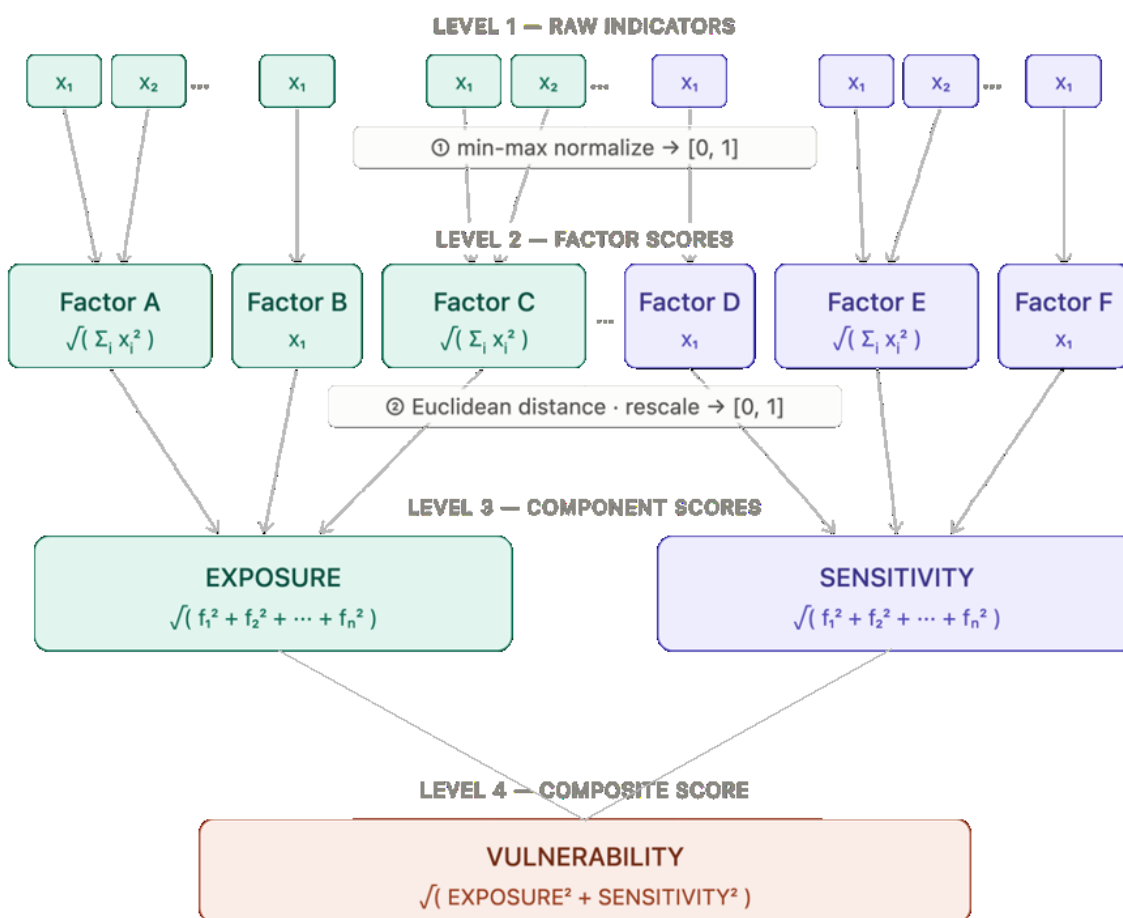


Figure 1. Methodological overview of how values are aggregated at each level to calculate the final vulnerability index, a relative metric.

4. Exposure Indicators

Exposure indicators capture projected future change under climate scenarios. Most climate-based indicators are derived from statistically downscaled global climate model projections representing conditions expected at mid-century (2035-2065) relative to a historical baseline. For some indicators, the underlying projection datasets were available only as precomputed products, so in those cases the analysis used the historical and future periods defined by the source dataset. These

time ranges are specified in the indicator descriptions below. Our approach uses a multi-model ensemble to allow us to characterize the extreme end of the projected indicator for scenario planning (accounting for outliers) weighted by the uncertainty in that projected value (Knutti et al. 2010).

4.1 Demand Exposure Factor

Nearby Water Use Exposure

Changes in nearby water withdrawals provide an indicator of potential demand-side pressure on water availability. Increases in water use near a water supply may reflect greater competition for local or regional water resources, especially in areas where future climate conditions are also expected to reduce supply. This factor is intended to capture projected changes in human water demand in the county containing each water supply.

For this factor, each water supply is assigned to its intersecting county. Historical water withdrawals are calculated from USGS observed water use data, while projected future withdrawals are calculated from Warziniack et al. (2022) water demand projections. These projections are based on 20 climate futures, representing five GCMs crossed with four SSP/RCP futures: SSP1/RCP2.6, SSP2/RCP4.5, SSP3/RCP7.0, and SSP5/RCP8.5. Total withdrawals are summed across major water-use sectors, including aquaculture, domestic, industrial and mining, irrigated agriculture, livestock, and thermoelectric water use.

Projected change in water withdrawals is calculated as percent change in mean annual withdrawals between the historical and future periods. Historical withdrawals are averaged across 1985-2015, while future withdrawals are averaged across the mid-century period, 2035-2065, for each projected demand scenario. For each county, percent change is calculated as the difference between future and historical mean annual withdrawals divided by historical mean annual withdrawals.

For each water supply, the projected withdrawal indicator is summarized using the 90th percentile value across future demand scenarios. The 90th percentile represents a plausible high-demand scenario, consistent with a precautionary approach to evaluating future water supply stress. Higher projected increases in nearby withdrawals indicate greater potential exposure to future demand pressure.

In addition to the withdrawal change indicator, we calculate the percentage of future demand scenarios that project an increase in withdrawals. This metric is applied as a weight to the withdrawal change value, indicating how consistently the projections suggest increasing water demand near each water supply.

Data Source: Historical water withdrawals are from USGS observed water use data. Future water withdrawals are from Warziniack et al. (2022) projected water demand data. Water withdrawals are summarized at the county scale and linked to water supplies based on the county containing each supply location.

Supply and Demand Mismatch Exposure [Provisional]

This exposure factor captures projected change in the alignment between water availability and recreational water demand. Even where annual water availability does not decline substantially, a water supply may become more vulnerable if the seasonal timing of runoff or precipitation shifts away from the months when visitation, and therefore water demand, is highest. This indicator is designed to capture that timing mismatch rather than total annual water volume.

For each water supply, monthly supply is represented using the hydrologic signal most relevant to the source type. (Rainwater-harvesting supplies use precipitation, because precipitation is the direct input to the system. Groundwater, springs, rivers, streams, lakes, reservoirs, and other non-ocean supplies use runoff as the source-water availability signal. Ocean or seawater sources are assigned a value of zero because their supply is not treated as seasonally limited in the same way as precipitation- or runoff-fed systems.)

Monthly supply and visitation are each converted to annual shares, so each 12-month curve sums to 1. This scaling preserves the seasonal distribution of each curve while placing supply and demand on a comparable unitless scale. The timing mismatch score is calculated as the area under the curve (AUC) where monthly visitation share exceeds monthly supply share. Larger values indicate a stronger seasonal mismatch, meaning a greater portion of visitor demand occurs during months when the selected supply signal is relatively low.

Projected timing exposure is calculated as the change in the AUC between the historical period and the future period. Historical supply is paired with historical visitation, while future supply is paired with projected future visitation for the same GCM, SSP, and run where available. Future visitation is generated from a park-level temperature-visitation modeling approach outlined in Fisichelli et al. (2015), where temperature-sensitive parks use their fitted temperature-response model and non-temperature-sensitive parks carry forward their historical monthly visitation climatology. The final exposure indicator is summarized across the paired climate ensemble using the high-end projected AUC change, with the percentage of model combinations showing increased mismatch retained as a model-agreement diagnostic weighting factor.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Implemented	Change in Nearby Water Withdrawals	Warziniack et al. (2022) for historical and projected water withdrawals.	Percent change in mean annual county-level water withdrawals between the historical period, 1985-2015, and the future period, 2035-2065. Historical withdrawals are calculated from observed USGS water use data. Future withdrawals are calculated for each projected demand scenario from Warziniack et al. (2022). Total withdrawals are summed across major water-use sectors, and the 90th percentile value across future scenarios is used to represent a plausible high-demand outcome. This value is then weighted by the percentage of future demand scenarios projecting an increase in mean annual county-level water withdrawals as a	Standard: higher increase = more vulnerable

Status	Indicators	Data Source	Methodology	Vulnerability Direction
			measure of confidence in the projected risk.	
Provisional	Change in Monthly Timing Mismatch area under the curve (AUC)	LOCA2/MWBM historical and future runoff/precipitation projections (Pierce et al. 2023, Alder & USGS 2023) ; projected park-level visitation from PRISM-trained, LOCA2-driven visitation models; NPS Visitor Use Statistics	AUC is calculated for historical and future monthly supply-demand curves. Future supply and projected visitation are paired by GCM, SSP, and run where available. The indicator is calculated as future AUC minus historical AUC, summarized using the 90th percentile across paired climate/visitation projections to represent a plausible high-mismatch outcome. This value is then weighted by the percentage of future climate scenarios where future seasonal timing mismatch AUC is greater than historical AUC as a measure of confidence in the projected risk.	Standard: larger increase = more vulnerable

4.2 Drought Exposure Factor

Drought represents a direct climate stressor on water supply reliability because it reflects periods when atmospheric water demand exceeds available precipitation. For this factor, drought is characterized using the Standardized Precipitation-Evapotranspiration Index (SPEI), which is calculated from monthly climatic water balance: precipitation minus potential evapotranspiration (PET). Negative SPEI values indicate drier-than-normal conditions, while positive values indicate wetter-than-normal conditions. Because SPEI incorporates both precipitation and evaporative demand, it is well suited for evaluating drought risk under warming, when increases in PET can intensify drought even in areas where precipitation does not decline.

For each water supply, we calculate projected change in drought conditions as the difference between future and historical mean SPEI. SPEI is calculated at a 12-month accumulation scale, which captures longer-term moisture deficits that are most relevant to water supply systems than short-term meteorological drought. Projected changes in drought were calculated using monthly precipitation and PET values from LOCA2 climate projections processed through a SPEI drought workflow. Monthly climatic water balance was calculated as precipitation minus PET, and SPEI was calculated for each GCM x SSP combination (n = 72). Mean SPEI was calculated for the historical

period, 1980-2010, and the mid-century period, 2035-2065. Change was then calculated as future mean SPEI minus historical mean SPEI. Ocean supplies are assigned a projected change of 0 because drought is not expected to change ocean quantity in the same way as precipitation- or runoff-fed supplies.

Projected drought exposure is summarized using the 10th percentile value across all climate projections. More negative SPEI change values indicate a shift toward drier future conditions and therefore greater vulnerability. In addition to the projected change indicator, we calculate the percentage of climate model combinations that project a decrease in SPEI. This model agreement metric indicates how consistently the ensemble projects drying at a given water supply.

Infinite SPEI values can occur when projected climatic water-balance values extend beyond the fitted historical distribution. Because extreme SPEI values derived through extrapolation are directionally meaningful but highly uncertain in magnitude, infinite values were treated as censored extremes and capped at ± 3 for numerical aggregation as recommended by Stagge et al. (2015).

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Implemented	Change in SPEI (Water Deficit)	LOCA2 statistically downscaled climate projections at ~6 km resolution (Pierce et al. 2023).	Change in mean SPEI between historical and future periods, using the 10th percentile (p10) value across climate projections. SPEI is calculated from monthly climatic water balance, defined as precipitation minus PET, using a 12-month accumulation scale. Mean SPEI is calculated for each historical and future period by GCM x SSP-run combination, and the model combination closest to the p10 of the SPEI-change distribution is selected. This metric is then weighted by the percentage of climate projections predicting a decrease in SPEI as a measure of confidence in the projected risk.	Inverted: negative values (drier conditions) = more vulnerable

4.3 Coastal Impacts Exposure Factor [Semi-Provisional]

Sea level rise (SLR) threatens coastal freshwater supplies through several related but distinct processes. Rising mean water levels can directly inundate wells, intakes, treatment facilities, and the surrounding landscape, shift the saltwater–freshwater interface inland within coastal aquifers, and increase the baseline water level on which storm surge occurs (Nicholls & Cazenave 2010,

Ferguson & Gleeson 2012). Climate-driven changes in storm characteristics may also alter extreme storm-surge levels independently of mean sea-level change (Muis et al. 2023). Global mean sea level is projected to rise approximately 0.3–1.0 m by 2100 under moderate- to high-emissions scenarios, although the magnitude and timing of relative sea-level rise vary regionally because of ocean dynamics and vertical land movement (Sweet et al. 2022, Kopp et al. 2014).

The coastal impacts exposure factor represents three pathways: expansion of inundation due to sea level rise around a water supply, projected inland migration of the saltwater–freshwater interface, and changes in extreme storm-surge levels. NOAA’s Continually Updated Shoreline Product (CUSP) provides the common coastline reference used to identify coastal water supplies and calculate distance to the coast (NOAA NGS 2020). Water supplies farther than 50 km from the CUSP shoreline are considered outside the coastal analysis area and receive NA for the coastal indicators to which the threshold is applied. Saltwater intrusion and storm surge are weighted by their distance to coastline, which is calculated as the Euclidean distance from each water supply point to the nearest CUSP shoreline feature.

Each indicator is calculated at the individual water-supply location and standardized so that larger values consistently represent greater exposure. The standardized indicators are then combined using the project’s Euclidean-distance framework. Where an indicator is not spatially applicable (for example, at an inland location outside the coastal analysis area or a location beyond the coverage of the underlying dataset) it remains NA. Because the Euclidean calculation removes missing values when summing the squared indicator scores, an unavailable indicator contributes zero to the calculated distance rather than being treated as an observed low-exposure value. Consequently, factor scores for sites with missing indicators are based on fewer components and should be interpreted accordingly.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Implemented	Projected SLR Inundation	NOAA Sea Level Rise Viewer inundation extent rasters matched to Interagency Sea Level Rise Task Force (ITF) 83 rd -percentile 2050 sea level projections (Sweet et al. 2022).	Percentage-point change in inundated area within the 4 km water supply buffer between the present-day surface and the NOAA inundation depth increment nearest to each site’s ITF 83 rd percentile SLR projection.	Standard: higher inundation change = more vulnerable
Implemented	Projected Saltwater Intrusion	Adams et al. (2024) global saltwater intrusion migration dataset, combined with distance to coast.	Each water supply is assigned the projected saltwater-freshwater interface migration due to recharge change and sea level rise out to 2100 from the nearest available coastal watershed centroid in the dataset. This value is then divided by the water supply’s distance to coast to provide a unitless ratio of the fraction of the corridor intruded. These ratios are	Standard: higher ratio (greater fraction of the corridor intruded) = more vulnerable

Status	Indicators	Data Source	Methodology	Vulnerability Direction
			capped at 1 since any intrusion beyond the water supply point provides no new information on the severity of the intrusion.	
Provisional	Projected Storm Surge	Muis et al. (2023) 100-year return period (RP100) storm surge projections, combined with distance to coast.	Each water supply is assigned the worst case (largest) change in RP100 storm surge level across the 4-model ensemble at its nearest surge station. This value is divided by the water supply's distance to the coast to weight exposure by coastal proximity.	Standard: larger surge increase = more vulnerable

4.4 Temperature Exposure Factor [Provisional]

Increasing air temperatures can contribute to water-quality degradation by warming surface water, accelerating chemical and biological reaction rates, reducing dissolved oxygen, and creating conditions favorable to pathogens and harmful algal blooms (Whitehead et al., 2009; Delpla et al., 2009). Groundwater temperatures also respond to atmospheric warming, although changes are generally slower and moderated by aquifer depth and groundwater residence time. Groundwater warming can nevertheless alter subsurface biogeochemical processes, groundwater-dependent ecosystems, and groundwater quality (Benz et al., 2024).

For this factor, projected change in mean annual air temperature is used as a spatially consistent indicator of the warming pressure experienced by each water supply. Air temperature does not represent the actual temperature of the source water, which is unavailable nationally, but it provides a consistently modeled climate forcing that can be calculated for every water supply as a surrogate. Its effect is expected to be most direct for rivers, streams, lakes, reservoirs, and rainwater supplies and more indirect for groundwater and spring supplies.

Projected temperature changes are calculated from monthly temperature values for the available GCMs and three SSPs: SSP245, SSP370, and SSP585. For each GCM x SSP combination, monthly temperatures are aggregated to annual mean temperature and spatially averaged across a 3x3 grid cell window surrounding each water supply. Change is calculated as the future mean annual temperature minus the historical mean annual temperature and is expressed in degrees Celsius. For each supply, projected temperature change is represented by the 90th percentile across the available climate model combinations. The 90th percentile represents a plausible warm-end projection and is used because greater warming is associated with greater potential for temperature-related water-quality degradation.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
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Provisional	Change in Mean Annual Temperature	LOCA2 statistically downscaled temperature projections at ~6 km resolution (Pierce et al. 2023), processed for the water-supply locations using the Jay Alder LOCA2/MWBM data products (Alder & USGS 2023)	Difference in mean annual temperature between the historical and future periods, using the 90th percentile value across the GCM–SSP projections. Values are spatially averaged across the grid cells surrounding each water supply, annual means are calculated for each GCM–SSP combination, and the modeled value closest to the p90 of the temperature-change distribution is selected.	Standard: greater projected warming = more vulnerable
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4.5 Water Supply Exposure Factor [Semi-Provisional]

Changes in precipitation and subsequent runoff represent some of the most direct climate impacts on water supply (Milly et al., 2005). Many CONUS regions, particularly the Southwest, are projected to experience declines in mean annual runoff under warming because of changes in precipitation and increased evapotranspiration (Seager et al., 2007; Hagemann et al., 2013). Climate change can also alter the seasonal timing of water supply by changing precipitation patterns, snow accumulation, and snowmelt. Shifts toward earlier or later runoff may affect water supply reliability even where total annual water availability changes relatively little (Stewart et al., 2005; Dudley et al., 2017).

This factor therefore evaluates two complementary aspects of projected water supply: the total annual amount of water and the timing of that water within the year. For rainwater-based supplies, precipitation is used because it is the direct input to the system. For groundwater, springs, and surface-water supplies, runoff is used because it better represents the hydrologic input most directly connected to those systems. Ocean supplies are assigned an amount change and timing change of 0 because ocean water availability is not expected to change in quantity or seasonal timing in the same way as precipitation- or runoff-fed supplies.

Projected change in water supply amount is calculated as the percent difference in mean annual precipitation or runoff between the historical and future periods. The 10th-percentile value across the climate-model ensemble is used to represent a plausible low water supply outcome. This change is paired with the percentage of climate-model combinations that project a decrease in the relevant water supply variable. Higher agreement indicates greater confidence that the projected decline is consistent across the ensemble (Tebaldi & Knutti, 2007).

Projected change in water supply timing is measured using the circular center of timing, a water-weighted measure of when precipitation or runoff is concentrated within the water year. Monthly values are placed on a circular October–September calendar, avoiding an artificial discontinuity between the end and beginning of the water year. A center of timing is calculated for each complete water year and climate-model combination, and timing change is expressed as the number of days between the historical and future centers.

Because both earlier and later shifts may disrupt established seasonal water supply patterns, timing exposure is based on the absolute magnitude of the projected shift. The direction supported by the greatest number of model combinations is selected. The 10th-percentile signed change is used when earlier timing is dominant, and the 90th-percentile signed change is used when later timing is dominant. If the ensemble is divided equally between earlier and later shifts, the direction with the larger absolute tail change is selected. Timing model agreement is the percentage of valid climate projections supporting the selected direction.

Projected changes in precipitation and runoff are calculated using monthly values from 27 GCMs and three SSPs: SSP2-4.5, SSP3-7.0, and SSP5-8.5. Together, these produce 72 GCM × SSP combinations. Water supply amount and timing are calculated for the mid-century period, 2035–2065, relative to the historical period, 1980–2010. Each future SSP simulation is matched to the corresponding historical simulation from the same GCM and run to reduce the influence of differences in baseline conditions among models. We used LOCA2 (Pierce et al., 2023) statistically downscaled climate projections at ~6 km resolution, processed through the Monthly Water Balance Model (MWB) developed by Jay Alder (Alder & USGS 2023), a Thornthwaite-Mather monthly water balance approach (Thornthwaite & Mather, 1957). The hindcast configuration uses each GCM’s historical run matched to its future SSP run to control for inter-model bias.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Implemented	Change in Water Supply Amount	LOCA2 (Pierce et al., 2023) statistically downscaled climate projections at ~6 km resolution, processed through the Monthly Water Balance Model (MWB) developed by Jay Alder (Alder & USGS 2023)	Percent change in mean annual water availability between historical and future periods, using the 10 th percentile (p10) value across all 72 climate projections. Values are spatially averaged in a 3x3 grid cell window around the water supply; annual totals computed per GCM+SSP model; model closest to the p10 of the percent-change distribution is selected. This value is then weighted by the percentage of climate projections predicting a decrease in mean annual water supply as a measure of confidence in the predicted risk.	Inverted: negative change (decreased water availability) = more vulnerable
Provisional	Change in Water Supply Timing		Change in the circular center of water source timing between the historical and future periods, expressed as the number of days shifted. Monthly runoff/precipitation is spatially averaged across the grid-cell neighborhood surrounding each water supply, and a runoff-/precipitation-weighted center of timing is calculated for each complete water year and GCM x SSP. The direction supported by the	Standard: higher fraction = more vulnerable

Status	Indicators	Data Source	Methodology	Vulnerability Direction
			most model realizations is selected: the 10th-percentile value is used when earlier timing is dominant and the 90th-percentile value when later timing is dominant. The absolute magnitude of the selected shift is used as the indicator. This value is then multiplied by the percentage of climate projections predicting the selected direction of timing change as a measure of confidence in the predicted timing shift.	

4.6 Wildfire Exposure Factor

Climate change is a primary driver of increasing wildfire activity across the western U.S. (Westerling et al., 2006; Abatzoglou & Williams, 2016). Wildfires pose risks to water supply through post-fire debris flows, ash introduction, and watershed destabilization (Bladon et al. 2014, Robinne et al. 2018).

Projected change in wildfire is derived from Gao et al. (2021) pyrome-level fire probability projections based on an ensemble of statistically downscaled Global Climate Models (CMIP5) in combination with the Physical Chemistry Fire Frequency Model (PC2FM) (Guyette et al. 2012) for 20 GCMs and two RCP scenarios (RCP 4.5 and RCP 8.5). Pyromes are zones of relatively homogeneous contemporary fire regime, delineated by clustering historical wildfire activity statistics. (Short et al. 2020). We selected the 90th percentile of the 40 climate projections as a plausible high-risk fire scenario. We additionally weighted the projected change by the percentage of climate projections (GCM x RCP combinations) predicting an increase in fire probability. This is an added measure of confidence in the projected risk.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Implemented	Change in Fire Probability	Gao et al. (2021) pyrome-level fire probability projections	Percent change in annual fire probability from baseline (1971-2000) to mid-century (2040-2069) at the 90th percentile (p90) across all 40 climate projections for the water supply's pyrome. We then weighted this metric by the percentage of climate projections predicting an increase in fire probability as an added measure of	Standard: higher projected increase = more vulnerable

Status	Indicators	Data Source	Methodology	Vulnerability Direction
			confidence in the projected risk.	

5. Sensitivity Indicators

Sensitivity indicators reflect current conditions that make a water supply more or less susceptible to climate impacts. They include physical hazard conditions, operational characteristics, demand pressures, and historical trends. The distinction between exposure (future-projected) and sensitivity (current-state) allows managers to understand the threat level and current susceptibility independently.

5.1 Demand Sensitivity Factor [Semi-Provisional]

Increasing visitation and regional water use competition create demand pressures that compound supply-side risks from climate change (Fisichelli et al. 2015). Increased regional water withdrawals reduce availability of shared water resources that some park systems depend on (Vörösmarty et al. 2010 Hoekstra et al. 2012).

The supply-demand mismatch indicator represents the historical degree to which a water supply's monthly water availability is mismatched with monthly park visitor demand. This indicator reflects an existing sensitivity of the system: supplies whose historically low-availability months coincide with peak visitation may be more susceptible to future water stress, even before considering projected climate change.

For each water supply, a historical monthly water climatology is calculated over the baseline period. Rainwater-harvesting systems use precipitation as the supply signal, while groundwater, springs, surface water, lakes, reservoirs, and other non-ocean supplies use runoff (ocean or seawater sources are assigned a value of zero, reflecting the assumption that these sources do not have a seasonal supply pulse comparable to precipitation- or runoff-fed systems). Historical monthly visitation is calculated for each park unit from NPS recreation visitation records over the same baseline period.

Both the monthly supply profile and monthly visitation profile are converted to annual shares by dividing each monthly value by its annual total. The indicator then calculates the "area under the curve (AUC)" where the visitation share exceeds the supply share across the 12-month seasonal cycle. A value near zero indicates that seasonal supply availability generally aligns with, or exceeds, seasonal visitor demand. Higher values indicate greater historical mismatch between when water is most available and when visitor demand is highest.

Because both curves are scaled to sum to 1, the score does not indicate whether total annual supply is sufficient to meet total annual demand. Instead, it identifies systems where the monthly pattern of demand is poorly aligned with the pattern of source-water availability. Absolute changes in annual runoff, precipitation, regional water use, and visitation magnitude are evaluated through separate indicators.

Status	Indicator	Data Source	Derivation	Vulnerability Direction
Implemented	Historic Visitation Trend	NPS Public Use Statistics (1990-2024)	Sen's slope (Sen, 1968) of annual recreation visitor counts, scaled as annual percent change relative to each park's long-term mean visitation. Significance assessed with the Mann-Kendall trend test (Mann, 1945; Kendall, 1975), a non-parametric method robust to non-normality and autocorrelation in time series. Parks with fewer than 3 years of data receive NA.	Standard: increasing trend = more pressure = more vulnerable
Implemented	Competition (Nearby Water Use)	USGS County-Level Water Withdrawal Data (1985-2015; USFS RPA compilation)	Sen's slope of total county-level water withdrawals across all major sectors (domestic, industrial, irrigated agriculture, aquaculture, livestock, thermoelectric). Each water supply assigned to nearest county via spatial join. Withdrawals normalized by county area (Mgal/day/km ²) before trend estimation.	Standard: increasing regional demand = more competition = more vulnerable
Provisional	Historical Supply and Demand Timing Mismatch AUC	MWBM historical runoff and precipitation data; NPS Stats historical recreation visitation data	Monthly water availability and monthly visitation are summarized over the historical period, 1984-2014. Water availability is represented by runoff (or precipitation). Supply and visitation are each converted to monthly shares of their annual total. The indicator is calculated as the area under the curve where monthly visitation share exceeds monthly water supply share.	Standard: higher projected increase = more vulnerable

5.2 Drought Sensitivity Factor

Historical drought trends provide empirical evidence of whether a water supply has already experienced increasing water deficits over time. A long-term shift towards more severe drought conditions suggests that the system may be more sensitive to future climate change because water availability has already been declining relative to the atmospheric water demand. This factor quantifies the historical change in drought severity using the Standardized Precipitation-Evapotranspiration Index (SPEI), which integrates both precipitation and evaporative demand to represent changes in climatic water balance.

Monthly SPEI values are calculated from historical LOCA2 precipitation and potential evapotranspiration (PET) data using a 12-month accumulation period to capture persistent hydrologic drought conditions relevant to water supply systems. Annual mean SPEI values are then calculated for the historical period (1980-2010) after spatially averaging data with a 3 x 3 grid cell window surrounding each water supply. Sen's slope is used to quantify the long-term trend in annual SPEI. Negative trends indicate increasingly severe drought conditions over time and therefore greater sensitivity to future climate change, while positive trends indicate decreasing drought sensitivity.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Implemented	Historical Drought Trend	LOCA2 statistically downscaled climate projections at ~6 km resolution (Pierce et al. 2023).	Sen's slope of annual mean 12-month SPEI across the historical period (1980-2010). Monthly SPEI is calculated from LOCA2 precipitation and potential evapotranspiration using a 12-month accumulation period. Climate variables are spatially averaged within a 3 x 3 grid cell window around each water supply before calculating annual mean SPEI and the historical trend.	INVERTED: negative trend (increasing drought severity) = more vulnerable

5.3 Flood Sensitivity Factor

Flooding poses direct risks to water supply infrastructure and can cause contamination through overland flow of pathogens, chemicals, and sediment. Climate change is projected to intensify extreme precipitation and associated flooding in many regions (Milly et al. 2002, Kunkel et al. 2013, Ward et al. 2015).

Status	Indicator	Data Source	Derivation	Vulnerability Direction
Implemented	Current Flood Risk	FEMA National Flood Hazard Layer (NFHL), queried via ArcGIS REST API	Percentage of the 4 km water supply buffer within FEMA Special Flood Hazard Areas (zones designated A or V), representing a ≥1% annual chance of flooding. Flood zone polygons queried via bounding box of each buffer, intersected with the buffer polygon, and area calculated. Sites with no FEMA data coverage receive NA; sites with FEMA data but no Special Flood Hazard Areas within the buffer receive 0%.	Standard: higher % in flood zone = more vulnerable

5.4 Sea Level Rise Sensitivity Factor

Current coastal inundation provides a measure of a water supply's existing physical susceptibility to sea level rise. Areas that are already low-lying and inundated under present-day coastal water

levels have less elevation and spatial separation from marine waters, making water supplies in these settings more sensitive to additional increases in sea level.

We quantify current inundation using the 0 ft sea level rise inundation extent from NOAA's Sea Level Rise Viewer. For each water supply, the percentage of the surrounding 4 km buffer that falls within the present-day inundation extent is calculated. A larger currently inundated area indicates that a greater proportion of the landscape surrounding the water supply is already susceptible to marine inundation and is therefore assigned greater sensitivity. Water supplies more than 50 km from the shoreline are considered outside the coastal analysis area and are assigned NA.

This sensitivity indicator complements the projected inundation exposure indicator described in Section 4.3. Current inundation describes the baseline susceptibility of the landscape, whereas exposure represents the additional expansion of inundation expected under future sea level rise. Together, the two indicators distinguish between water supplies that are already situated in inundation-prone coastal environments and the magnitude of additional coastal change they may experience in the future.

Status	Indicator	Data Source	Methodology	Vulnerability Direction
Implemented	Current Inundation	NOAA Sea Level Rise Viewer inundation extent rasters with 0ft of SLR.	Percentage of the 4km water supply buffer that is inundated under present day (0ft SLR) conditions. Sites >50km from the shore are assigned NA and excluded from standardization.	Standard: Higher percentage currently inundated = more vulnerable

5.5 Source Type Sensitivity Factor

Water-source type fundamentally shapes a supply's baseline sensitivity to climate variability. Surface-water supplies respond relatively quickly to changes in precipitation, runoff, and wildfire, whereas groundwater supplies generally buffer short-term variability but remain sensitive to sustained changes in recharge and withdrawals (Green et al. 2011, Taylor et al. 2013).

For groundwater-fed supplies, depth to the water table indicates the degree of connection between the aquifer and conditions at the land surface. Shallow groundwater has a shorter intervening flow path and can therefore respond more rapidly to recharge deficits and surface-derived contamination. Greater depths generally provide more separation and buffering from these surface influences. Accordingly, shallower groundwater is assigned greater sensitivity.

For groundwater- and spring-sourced supplies, depth to groundwater is estimated first from the well depth recorded in the Water Supply Database (WSD), when available. Total well depth is not a direct measurement of static water level: wells are typically completed below the water table to accommodate seasonal drawdown and keep the pump intake submerged (Sterrett 2007). Recorded well depth is therefore treated as an upper bound on water-table depth. Comparisons of WSD records containing both well depth and observed static water level showed that subtracting a fixed depth from the well record continued to overestimate water-table depth and that the discrepancy increased with well depth. Based on this comparison, water-table depth is approximated as 80% of the recorded total well depth.

Where recorded well depth is unavailable, depth to groundwater is obtained from the ~30 m CONUS water-table-depth dataset developed by Ma et al. (2026). This dataset was produced using a 300-tree random-forest model trained on more than one million groundwater observations; the median prediction across the decision-tree ensemble is used here.

Estimated depths greater than 200 m are capped at 200 m before scoring. The threshold was selected from the upper tail of groundwater depths represented in the water-supply dataset and is treated as an analytical saturation point rather than a physical maximum. A national analysis of approximately 5.3 million wells used depths of 200 ± 100 m to distinguish very deep wells that are increasingly likely to access old or fossil groundwater (GebreEgziabher et al. 2022). The cap therefore prevents a small number of extreme values from disproportionately affecting the indicator while retaining the important distinction between shallow, intermediate, and very deep groundwater sources.

All other source types are assigned a depth of 0 m because they represent surface-water supplies that are directly exposed to changing surface-climate conditions.

Status	Indicator	Data Source	Methodology	Vulnerability Direction
Implemented	Depth to Water Table	NPS Water Supply Database (recorded well depth); Ma et al. (2026) machine-learning water table depth model (~30m resolution, CONUS)	For groundwater- and spring-sourced supplies, depth to water is estimated as 80% of recorded well depth. Where well depth is unavailable, the modeled value at the supply location (or the nearest valid value within an expanding 500 m to 50 km search radius) is used. Depths greater than 200 m are capped at 200 m. All other source types are assigned a depth of 0 m.	Inverted: Shallower depth to water = more vulnerable

5.6 Temperature Sensitivity Factor [Provisional]

The sensitivity of a water supply to additional warming depends partly on its existing thermal conditions and whether it is already experiencing a long-term warming trend. Supplies in warmer environments may be closer to temperature thresholds associated with reduced dissolved oxygen, accelerated chemical and biological activity, pathogen growth, or harmful algal blooms. A supply that has already experienced sustained historical warming may also have less capacity to absorb additional temperature increases without water-quality consequences.

For this factor, the historical trend indicates whether warming has already been occurring. This indicator describes ambient climate conditions rather than measured source-water temperature because nationally consistent water-temperature observations are not available for all NPS water supplies.

The temperature trend is calculated using Sen's slope, a nonparametric method that estimates the median rate of change through time and is relatively insensitive to individual extreme years. Annual mean temperature per grid cell is spatially averaged across a 3x3 grid cell window surrounding each water supply.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Provisional	Historical Temperature Trend	LOCA2 statistically downscaled temperature projections at ~6 km resolution (Pierce et al. 2023).	Sen's slope of annual mean temperature across the historical period, expressed in degrees Celsius per year. Annual mean temperatures are calculated for each grid cell and then spatially averaged around each water supply before the trend is estimated.	Standard: a greater positive warming trend = more vulnerable

5.7 Water Supply Trend Sensitivity Factor

Historical precipitation and subsequent runoff trends provide empirical evidence of whether a water supply system is already experiencing declining water availability. A negative historical trend indicates conditions are already deteriorating, suggesting greater sensitivity to additional projected decreases (Milly et al. 2005, Dettinger & Cayan 1995).

For this factor, we use either projected change in runoff or projected change in precipitation, depending on the water supply type. For rainwater-based supplies, we use precipitation because it is the direct input to the system. For groundwater, springs, and surface water supplies, we use runoff because it better represents the hydrologic input most directly connected to those systems. Ocean supplies are assigned a projected change of zero because ocean water availability is not expected to change in quantity in the same way as precipitation- or runoff-fed supplies.

Status	Indicators	Data Source	Methodology	Vulnerability Direction
Implemented	Historical Water Supply Trend	LOCA2 MWBM (Alder & USGS 2023)	Sen's slope of annual water supply across the historical period (1980-2010). The selected variable depends on water supply type: precipitation is used for rainwater-based supplies, runoff is used for groundwater, springs, and surface water supplies, and ocean supplies are assigned a trend of 0 because changes in supply quantity are not applicable. Annual values are calculated from the LOCA2 MWBM historical simulation and spatially averaged in a 3x3 grid	INVERTED: negative trend (declining water availability) = more vulnerable

			cell window around the water supply grid cell.	
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5.8 Wildfire Sensitivity Factor

Current wildfire hazard potential reflects the inherent flammability of the landscape surrounding each water supply, independent of future climate projections. High-hazard areas are at elevated risk of post-fire water quality degradation through ash introduction, hydrophobic soil layers increasing erosion, and burned riparian vegetation reducing stream bank stability (Bladon et al. 2014, Murphy et al. 2018, Robinne et al. 2018).

Status	Indicator	Data Source	Methodology	Vulnerability Direction
Implemented	Current Wildfire Hazard	USFS WHP2023 (Dillon et al., 2023); 270 m categorical raster, 5 classes	Area-weighted percentage of the 4 km water supply buffer in WHP categories 4 (high) or 5 (very high). A binary raster is derived; terra::extract(weights=TRUE, exact=TRUE) computes the weighted fraction in the high/very-high classes.	Standard: higher % high-hazard = more vulnerable

6. Priority Classification

Priority designations are based on both the overall vulnerability score and hazard-specific flags. For the overall vulnerability rank, water supplies in the top 25% of vulnerability scores for the current comparison group (i.e., $\geq$ 75th percentile) are flagged as high priority. This means that when viewing supplies at the national level, a high priority water supply has an overall vulnerability score higher than 75% of all CONUS water supply systems, and when a user filters the dataset to either NPS Region or state it recalculates these percentiles and therefore the high priority designations. This allows for identification of high-priority systems within their regional context of interest.

Hazard flags are assigned when a factor score exceeds a normalized value of 0.75, identifying systems of high priority in reference to single-hazard exposure or sensitivity. As with the vulnerability index, within the app these flags are recalculated based on the user Region or state filters.

Flag	Trigger Condition
Drought	Drought (SPEI) sensitivity or exposure factor ≥ 0.75
Fire	Wildfire exposure factor ≥ 0.75 or wildfire sensitivity factor ≥ 0.75
Flood	Flood sensitivity or exposure factor ≥ 0.75
Sea Level Rise	SLR sensitivity or exposure factor ≥ 0.75

7. Contribution Calculations

For any selected water supply, the tool reports how much each indicator, factor and component contributes to that site's scores. Because aggregation uses Euclidean distance, the squared value of each input is its exact additive share of the parent's squared score: if a factor score F is calculated as $F = \sqrt{(\sum x_i^2)}$, then $\sum x_i^2 = F^2$, and each x_i^2 / F^2 is the fraction of that factor's squared score attributable to indicator i , with all shares summing to one. Contributions are therefore reported as squared shares. One consequence is that contribution percentages are not proportional to the indicator scores themselves: an indicator scoring twice as high as another contributes four times as much, which follows directly from the aggregation method described in Section 3.3.

The **Component Contribution** chart applies this at the factor level. Each factor's contribution is calculated as $100 \times f_i^2 / \sum f_j^2$, where f_i is the site's factor score and the summation runs over all scored factors within Exposure or Sensitivity. Factors that are not applicable to the site's source type or location (e.g., coastal indicators) and factors with no available data are excluded from the denominator and displayed as unscored rather than as zero, so reported percentages are shares of the score that was actually calculated, not of a hypothetical complete score. Hovering over a multi-indicator factor applies the same squared-share calculation one level down, across that factor's normalized indicators.

The **All Levels Contribution** chart expresses every level of the hierarchy as a share of the site's overall Vulnerability score, obtained by chaining squared shares multiplicatively from the top down. Because component scores are retained as raw Euclidean distances rather than re-normalized, Exposure and Sensitivity partition Vulnerability exactly: $E^2 + S^2 = V^2$. Vulnerability is therefore set to 100%, with Exposure and Sensitivity receiving $E^2 / (E^2 + S^2)$ and $S^2 / (E^2 + S^2)$, and each factor receiving its within-component squared share multiplied by its component's share of Vulnerability. Values at these levels sum exactly to their parent, and the width of each box is proportional to its contribution. Indicator-level shares within the two multi-indicator factors are calculated the same way, as $x_i^2 / \sum x_j^2$ across that factor's normalized indicators, but because those factor scores are min-max re-normalized after aggregation (Section 3.3), these shares describe the internal composition of the factor rather than an exact continuation of the decomposition to the Vulnerability level. Only scored items are drawn, with N/A and missing entries reported in the Component Contribution chart instead.

All contribution percentages are relative rather than absolute. They describe the composition of a site's score, not its severity, so a factor can account for a large share of a low overall score. Because min-max normalization is recalculated within the active comparison group, contributions are also scope-dependent and will change when the user filters to a region or state. And because unscored factors are dropped from the denominator, sites with substantial missing data will show inflated contributions for the factors that remain.

8. Limitations and Caveats

8.1 Point Representation and Buffer Distance

Water supply systems are represented as point locations, sometimes with an additional spatial buffer. Four indicators - current flood risk, current wildfire hazard, current coastal inundation, and projected sea level rise inundation - summarize conditions within a 4 km buffer surrounding each water supply. Climate and hydrologic indicators derived from LOCA2 and the Monthly Water Balance Model (MWBM) are instead generally summarized across a 3x3 grid-cell neighborhood centered on the water supply location. They do not accurately capture the full contributing watershed for surface water systems or the recharge zone for groundwater systems. Additionally, numerous records lack precise source coordinates and are assigned park unit centroids, introducing positional uncertainty that propagates into all indicator calculations.

8.2 Equal Weighting

All indicators, factors and components are treated as equally important in the composite vulnerability score. To test whether that choice materially affects results, we reproduced the uncertainty-weighting sensitivity analysis of Michalak et al. (2021, Appendix D) against this assessment's factors. Using all of our Implemented and Provisional indicators, each factor was assigned an estimated uncertainty rank from 1 (low) to 5 (high) based on its data source, reliance on modeled or projected conditions, and the relative uncertainty of the underlying variable. Uncertain factors (rank 4-5) were then re-weighted by 0.5 (down-weighting) and by 2.0 (up-weighting) and the vulnerability score recalculated.

Down-weighting the uncertain factors produced a Spearman rank correlation of 0.94 with the unweighted index, and up-weighting produced 0.91. Both substantially exceed the corresponding values reported by Michalak et al. (0.86 and 0.49, respectively), indicating that our overall vulnerability rankings (and our set of indicators they are based upon) are considerably less sensitive to weighting than those of the framework we adapted from. Up-weighting changes rankings modestly more than down-weighting, consistent with the fact that squaring during aggregation means doubling a factor's weight quadruples its contribution to the composite score, but the difference between the two directions is small.

Because these scenarios depend on one particular judgment about which factors count as uncertain, we ran an additional randomized check in which each of the uncertainty rank groups received an independent weight drawn from a uniform distribution between 0.5 and 2.0, repeated across 2,000 draws. The resulting rank correlations with the unweighted index had a median of 0.90, with 95 percent of draws at or above 0.82. Rankings are therefore stable across a wide range of arbitrary weighting schemes.

Rank correlation is not the operationally relevant measure. Under the two weighting scenarios, 124 and 146 of 1,410 water supplies changed priority status, which corresponds to roughly 18 to 21 percent of the high-priority list changing membership. Because sites are densely distributed near the 75th-percentile threshold, small changes in rank produce disproportionately many crossings of that boundary. Overall rankings should therefore be regarded as stable under plausible weighting assumptions, while individual priority designations for sites near the threshold should be treated as indicative rather than definitive, and read alongside the percentile rank rather than the priority flag alone.

Two limitations apply to this analysis. The uncertainty ranks are provisional structural assignments rather than an expert elicitation, and they should be revised through the same review process used for the indicators themselves. They are also strongly collinear with the Exposure-Sensitivity division, since all eight factors in the highest-ranked groups are Exposure factors built from climate

projections. This analysis is consequently better understood as a test of how much the balance between projected and observed conditions affects results than as a test of the uncertainty attached to any individual indicator. Equal weighting is retained for transparency and reproducibility and because no expert-elicited weighting scheme is currently available, but the index should not be described as weighting-invariant.

8.3 Relative Scoring

The vulnerability index is designed to identify relative differences among NPS water supplies, not to estimate an absolute probability of water-supply failure, infrastructure damage, contamination, or other impact. Indicators are standardized onto a common scale and combined to show where each supply falls relative to the other supplies included in the active comparison group. A high vulnerability score therefore means that a supply has comparatively high values across the indicators evaluated in this assessment; it does not indicate that a specific adverse outcome is certain or that a particular threshold of acceptable risk has been exceeded.

The comparison group is also important. Within the application, normalization and ranking are recalculated when users filter the dataset by NPS Region or state. A water supply can therefore have a different normalized score, percentile rank, priority designation, or hazard classification when evaluated nationally than when evaluated only against supplies in its region or state, even though the underlying raw environmental data for that supply have not changed. National rankings are most appropriate for identifying systems that stand out across the entire NPS portfolio, whereas regional and state rankings identify systems that stand out within a more geographically similar group.

Priority status is similarly relative. Systems at or above the 75th percentile of the selected comparison group are identified as priority systems, so the designation identifies the upper portion of the current score distribution rather than an externally established threshold separating “safe” from “unsafe” systems. Sites immediately above and below this threshold may have very similar vulnerability scores. Priority classifications should therefore be interpreted together with the underlying percentile rank, raw indicator values, individual factor scores, and available data rather than as a binary determination of vulnerability.

Finally, because the indicator set continues to include provisional methods and will expand as planned indicators are implemented, the relative distribution itself may change as the assessment evolves. Scores and rankings should therefore be understood as describing vulnerability relative to the water supplies, indicators, and methods included in a particular version of the assessment.

8.4 Climate Scenario Uncertainty

All future projections carry inherent uncertainty: (1) scenario uncertainty, (2) model uncertainty (structural differences among GCMs), and (3) internal climate variability (Knutti et al., 2010). Meanwhile, ensemble statistics should not be interpreted as formal probabilities or confidence intervals. For example, a finding that 80% of climate-model combinations project a decline does not mean there is an 80% probability that the decline will occur. Climate models are not completely independent, the ensemble does not represent all possible future climate states, and the included SSPs are scenarios rather than outcomes with assigned probabilities.

Climate uncertainty is also represented differently among indicators because the assessment integrates multiple existing projection products. The LOCA2/MWBM indicators primarily evaluate mid-century conditions for 2035-2065 relative to 1980-2010, whereas other datasets have their own ensembles, scenarios, baseline periods, and projection horizons. For example, wildfire exposure is based on 40 projections from 20 CMIP5 models and two RCPs and uses a 2040-2069

future period; projected water demand uses 20 climate and socioeconomic futures; storm-surge change is derived from a four-model ensemble extending to approximately mid-century; projected sea level inundation is matched to an 83rd-percentile 2050 sea level projection; and saltwater intrusion projections extend to 2100. These differences reflect the best nationally consistent datasets available for each hazard but mean that the exposure indicators do not represent an identical set of climate futures or a single common projection period.

The composite vulnerability score should therefore not be interpreted as the outcome of one internally consistent future climate scenario. Instead, it combines screening-level indicators describing plausible adverse changes across several climate-related pathways. The tool is intended to identify water supplies and hazards warranting closer investigation, while site-specific adaptation planning should evaluate the full range of locally relevant climate projections, hydrologic processes, infrastructure characteristics, and management constraints.

Appendix A: Water Balance Model Variable Explorer

The variables presented in the Water Balance Model Variable Explorer come from the U.S. Geological Survey Monthly Water Balance Model (MWB), which translates statistically downscaled meteorology into a simplified representation of the terrestrial water balance. The LOCA2 application is driven by monthly mean air temperature and precipitation from 27 CMIP6 global climate models under SSP2-4.5, SSP3-7.0, and SSP5-8.5 and provides monthly climate and hydrologic variables on an approximately 6 km grid for 1950–2100 (Alder, 2023).

The MWB is a monthly water-balance model implemented and evaluated across CONUS by Hostetler and Alder (2016). It uses a set of parameterizations and rules to partition water among snowpack, evapotranspiration, soil-water storage, and immediate and delayed runoff. At each grid cell, air temperature is used to partition precipitation into rain and snow and to control snow accumulation and melt. The model estimates potential evapotranspiration from air temperature and extraterrestrial radiation using the Oudin method, calculates actual evapotranspiration as a function of precipitation, available soil moisture, and potential evapotranspiration, and updates soil-water storage subject to a spatially varying water-holding capacity. A fixed fraction of precipitation is assigned directly to runoff, while water exceeding the soil-water-holding capacity can contribute to delayed runoff (Hostetler & Alder, 2016; Oudin et al., 2005).

The MWB is designed to represent first-order, climate-driven hydrologic changes consistently across a large ensemble rather than reproduce every process at an individual water supply. It operates at a monthly time step, generates runoff locally without routing it between grid cells or through a stream network, and does not explicitly represent streamflow, groundwater, reservoirs, diversions, or other water-management operations. The variables are therefore interpreted as hydroclimatic indicators rather than site-specific observations or forecasts of true water yield (Hostetler & Alder, 2016).

For each variable, monthly values are extracted from the 3x3 grid-cell neighborhood centered on each water supply and spatially averaged to represent local conditions while reducing sensitivity to any single grid cell. Monthly values are then aggregated to a water year summary appropriate to the variable: precipitation, PET, AET, and runoff are summed to annual totals; temperature and soil-moisture storage are averaged across the water year; and SWE is represented by the water year maximum, reflecting peak seasonal snow storage. Annual values are then averaged separately across the historical and future periods for each GCM x SSP combination. Each future projection is compared with its corresponding historical hindcast from the same climate model.

The explorer provides complementary views of both the modeled conditions and their projected change. Users can view the historical baseline (1980-2010), future raw values (2035-2065), and change from historical conditions. For precipitation, PET, AET, runoff, SWE, and soil-moisture storage, percent change supports comparison among locations with different baseline climates, while absolute change in millimeters provides information about the actual magnitude of the projected shift. Temperature change is expressed as a difference in degrees Celsius rather than a percentage. Raw historical and future values provide additional context and are especially important where baseline values are small, because percent changes can become very large or unstable near zero.

For each selected metric, the explorer summarizes the distribution across the climate-model ensemble using values from the models closest to the 10th percentile (p10), median (p50), and 90th percentile (p90). The minimum and maximum are also available as optional diagnostics. Together, these views allow users to evaluate the projected direction and magnitude of change, the conditions from which that change occurs, and the spread of outcomes across the climate-model ensemble.

Variable Descriptions

The explorer includes the two meteorological inputs, temperature and precipitation, and five MWBM-derived variables: snow water equivalent, potential evapotranspiration, actual evapotranspiration, runoff, and soil-moisture storage. Water quantities are expressed as an equivalent depth of water over the grid cell, typically in millimeters.

Temperature: Monthly mean near-surface air temperature from LOCA2. Temperature is a model input rather than an MWBM output. Within the MWBM, it helps determine whether precipitation is treated as rain or snow, controls snow accumulation and melt, and is used with extraterrestrial radiation to calculate potential evapotranspiration. It should not be interpreted as the temperature of the source water.

Precipitation: Monthly total precipitation from LOCA2, expressed as liquid-water equivalent. Precipitation is a model input and the primary source of water entering the MWBM. Depending on temperature and existing model conditions, it is apportioned among snowpack, evapotranspiration, soil-water storage, and immediate or delayed runoff.

Snow water equivalent (SWE): The liquid-water equivalent of the water stored in the modeled snowpack. SWE accumulates when precipitation is classified as snow and declines as snow melts. It represents the depth of water contained in the snowpack rather than the physical depth of the snow. Annual maximum SWE represents the greatest modeled snow-water storage reached during a water year.

Potential evapotranspiration (PET): The amount of evapotranspiration that could occur under the modeled atmospheric conditions before limitations from available water are applied. The MWBM calculates PET from monthly mean air temperature and extraterrestrial radiation using the Oudin method. Extraterrestrial radiation varies with latitude and time of year, while air temperature scales the resulting potential evaporation rate. PET represents potential atmospheric water demand rather than actual water loss and does not explicitly account for differences among vegetation, land-cover, or other surface types (Hostetler & Alder, 2016; Oudin et al., 2005).

Actual evapotranspiration (AET): The modeled amount of water transferred from the land surface to the atmosphere through evapotranspiration after accounting for water availability. In the MWBM, AET is calculated as a function of precipitation, available soil moisture, and PET. AET may approach PET when sufficient water is available but is lower when precipitation and soil-water storage cannot meet the modeled evaporative demand.

Runoff: Total monthly runoff generated within the grid cell. The MWBM separates runoff into immediate and delayed components. A spatially uniform direct-runoff fraction of 0.05 assigns 5% of monthly precipitation directly to runoff, bypassing soil-water storage and evapotranspiration. The remaining water is processed through the model's snow and soil-water accounting, with water exceeding the soil-water-holding capacity contributing to water surplus and, subsequently, delayed runoff. In dry locations where soil-water storage does not reach capacity, direct runoff may constitute most or all of the modeled runoff. Runoff is generated locally within each grid cell and is not routed through a stream network; it should not be interpreted as streamflow or discharge at a particular channel, gage, or water-supply intake.

Soil-moisture storage: The amount of water retained in the model's soil column at a given time. Storage is limited by a spatially varying soil-water-holding capacity derived from soil characteristics and an assumed rooting depth of 1 m. Soil-water storage is replenished by available precipitation and snowmelt and depleted through actual evapotranspiration. Once the soil-water-holding capacity is reached, additional water can contribute to water surplus and delayed runoff. This variable represents the model's conceptual soil-water storage and should not be interpreted as a direct measurement of volumetric soil moisture, groundwater storage, or water available throughout an entire watershed. e storage capacity is filled, additional water can contribute to surplus runoff. This term represents the modeled soil-water bucket and should not be interpreted as a direct measurement of volumetric soil moisture or groundwater storage

Appendix B: Data Sources Summary

Dataset	Variable(s)	Period / Coverage	Used For
NPS Water Supply Systems Database	Source type, treatment type, coordinates, park unit	Current; ~1,400 CONUS systems	Study unit definition; source type and treatment type indicators
LOCA2 Climate Projections (Pierce et al., 2023)	Temperature (~6 km)	Historical 1950-2018; future 2015-2100; multiple GCMs; SSP2-4.5, SSP3-7.0, SSP5-8.5	Supply and demand mismatch sensitivity indicator
PRISM	Temperature (~4 km)	Historical (1979-2025)	Supply and demand mismatch sensitivity indicator
LOCA2 Monthly Water Balance Model (Jay Alder, USGS)	Modeled and historical runoff, precipitation, SWE, PET	Historical 1980-2010 (hindcast); future 2035-2065; multiple GCMs; SSP2-4.5, SSP3-7.0, SSP5-8.5	Runoff, precipitation, SWE, PET; all MWBM-based exposure and sensitivity indicators

NOAA Sea Level Rise Viewer	Inundation extent rasters (reference + depth-increment scenarios)	Current and projected (2100); state-level coastal tiles	SLR exposure (projected) and SLR sensitivity (current inundation)
WHP2023 — Wildfire Hazard Potential (Dillon et al., 2023)	WHP categorical raster (5 classes; 270 m)	Current conditions; CONUS	Wildfire sensitivity indicator
Gao et al. (2021) Fire Probability Projections	Pyrome-level fire probability deltas (p10, p50, p90)	Mid-century projected; CONUS pyromes; RCP 4.5 & 8.5	Wildfire exposure indicator
FEMA National Flood Hazard Layer (NFHL)	Flood zone polygons (A & V designations)	Current regulatory; CONUS (coverage varies)	Flood sensitivity indicator
NPS Visitor Use Statistics	Annual recreation visitor counts by park unit	1990-2024	Visitation trend (demand sensitivity)
USGS County Water Withdrawals (USFS RPA compilation)	County sectoral water withdrawals (Mgal/day), historical and future	1985-2015 (5-year intervals); 2035-2065	Nearby water use trend (competition sensitivity); Future water use trend (competition exposure)
NPS Park Boundaries (NPS Land Resources Division, 2025)	Park unit boundary polygons	Current (retrieved August 2025)	Spatial reference; centroid imputation for supplies missing coordinates
NOAA CUSP (Continually Updated Shoreline Product)	Shoreline vector tiles	Current; CONUS coastline	SLR sensitivity — distance to shoreline
Adams et al. (2024) Saltwater Intrusion Migration Dataset	Projected saltwater-freshwater interface migration distance	Projected to 2100; CONUS coastal watersheds	SLR exposure — projected saltwater intrusion
Muis et al. (2023) Storm Surge Projections	RP100 storm surge level, 4-model ensemble	Current and projected to 2050; CONUS coastal tide/surge stations	SLR exposure — projected storm surge

Ma et al. (2026) Water Table Depth Dataset	Water table depth raster (~30 m)	Current; CONUS	Water source type sensitivity indicator
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